



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111 – Topic 6

Operational Amplifiers

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Topic 6 Content

This lecture covers:

- Operational amplifiers (Op Amp) as active elements
- Ideal and non-ideal Op Amp
- Mathematical operations using Op Amp
 - Inverting Op Amp
 - Non-inverting Op Amp
 - Summing Op Amp
 - Difference Op Amp
 - Integrator and Differentiator

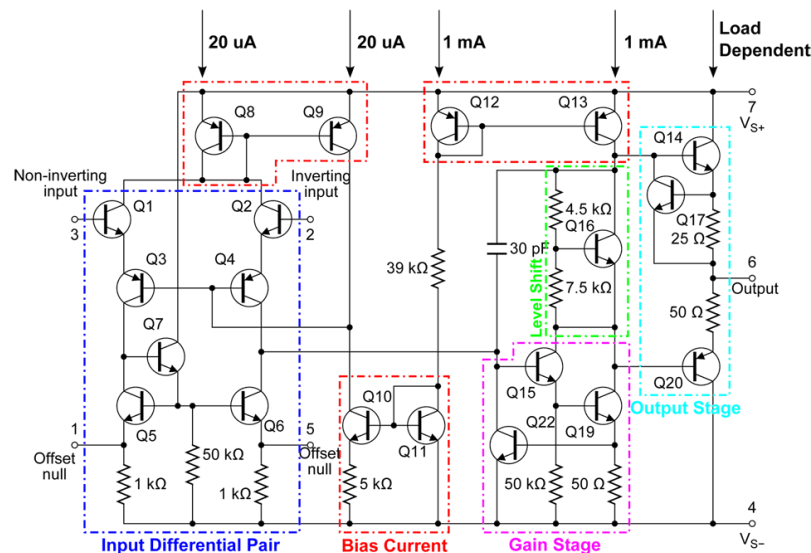
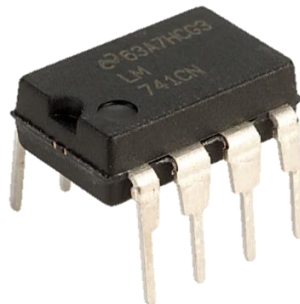
Corresponds to Chapter 5 and last part of Chapter 6 of your textbook

Addressed learning outcomes:

- *LO1*: Systematically analyse DC and AC electric circuits by deriving and solving equations using circuit laws and theorems.
- *LO4*: Apply concepts of DC and AC circuit analysis in circuits with ideal operational amplifiers.
- *LO6*: Evaluate relevant information to design simple engineering systems that use electrical circuits.

Operational Amplifiers

- Operational amplifiers or Op Amp are **active elements**.
- The Op Amp is an electronic device consisting of a complex arrangement of resistors and transistors.
- They are commercially available in integrated circuit packages (e.g. LM 741).
- An Op Amp circuit is designed to perform **mathematical operations** in **analog circuits** such as addition, subtraction, multiplication, division, differentiation, and integration when **resistors** and **capacitors** are connected to its terminals.

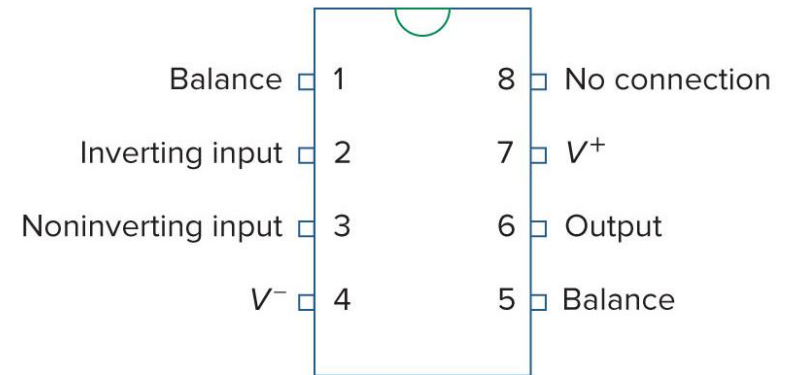
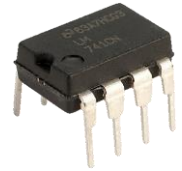


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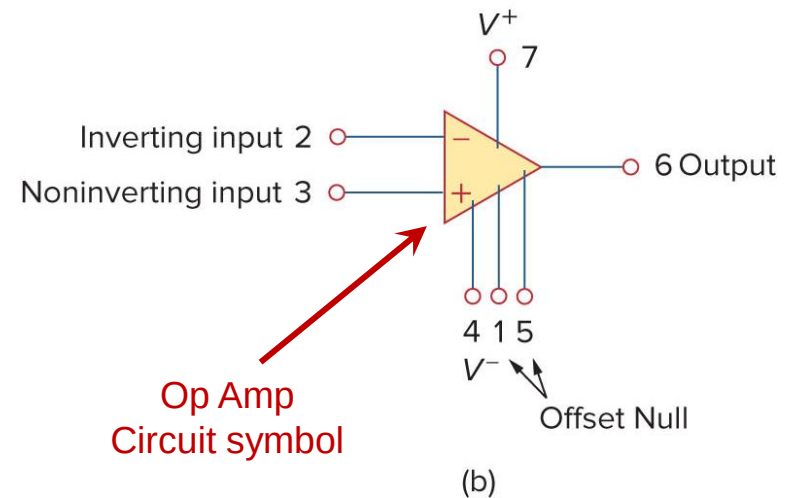
Operational Amplifiers

- A typical Op Amp has **eight/8 pins**. We are only interested in **five/5** of them:
 - **Inverting input** (pin 2)
 - **Non-inverting input** (pin 3)
 - **Output** (pin 6)
 - **Positive** power supply V^+ (pin 7)
 - **Negative** power supply V^- (pin 4)

- A signal applied to the **non-inverting** input will appear with the **same polarity** in the output.
- A signal applied to the **inverting** input will appear with the **opposite polarity** in the output.



(a)

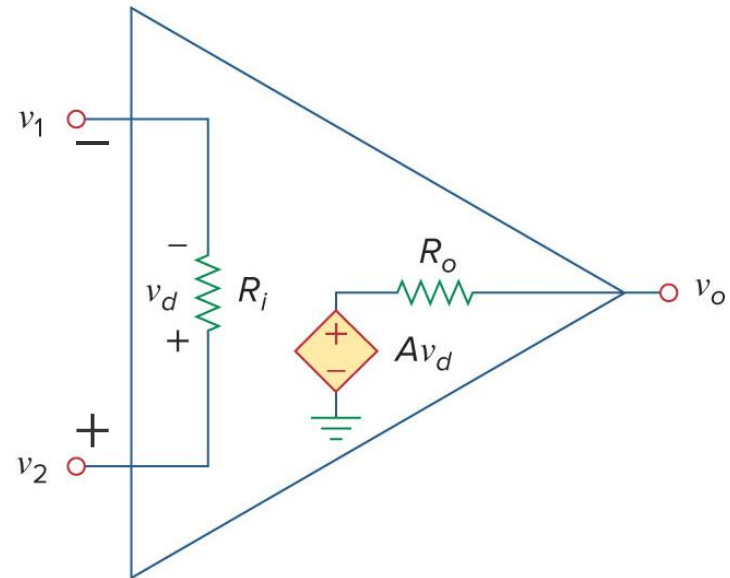


(b)

Equivalent circuit model

- An Op Amp is modelled from its input terminals by an **input resistor** R_i , and from its output terminal by a **voltage-controlled voltage source** in series with an **output resistor** R_o .
- The output voltage v_o is given by:

$$v_o = Av_d = A(v_2 - v_1)$$



Typical range for Op Amp parameters

Parameter	Typical range	Ideal value
Open-loop gain, A	10^5 to 10^8	∞
Input resistance R_i	10^5 to $10^{13} \Omega$	$\infty \Omega$
Output resistance R_o	10 to 100Ω	0Ω
Supply voltage, V_{cc}	5 to 24 V	

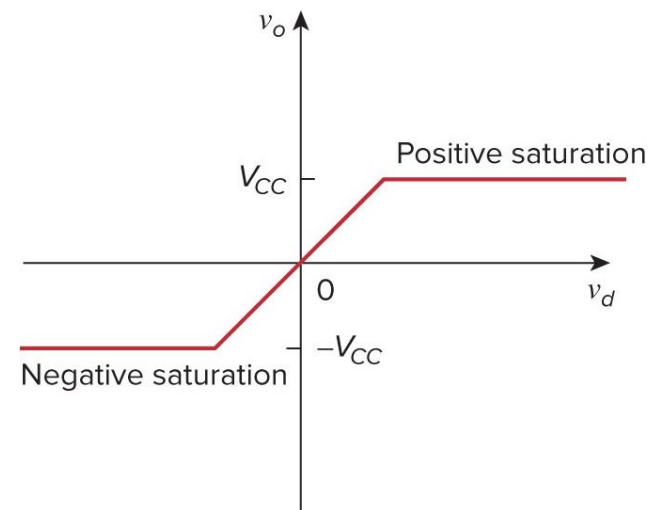
- A : **Open-loop voltage gain**
- v_d : **Differential** input voltage
- v_1 : **Inverting** terminal voltage to the **ground**
- v_2 : **Non-inverting** terminal voltage to the **ground**
- R_o : Thevenin equivalent resistance seen at the **output terminals**
- R_i : Thevenin equivalent resistance seen at the **input terminals**

Ideal vs non-ideal Op Amp

- In practice, the magnitude of output voltage v_o cannot exceed its power supply voltage $|V_{CC}|$.
- Depending on the differential input voltage in open-loop form, the Op Amp can operate in three modes:
 1. Positive saturation: $v_o = V_{CC}$
 2. Linear region: $-V_{CC} \leq v_o \leq V_{CC}$
 3. Negative saturation: $v_o = -V_{CC}$

We always operate Op Amp in its **linear region** throughout this course

$$-V_{CC} \leq v_o \leq V_{CC}$$



- Assuming **linear operation**, an **ideal Op Amp** has the following characteristics:
 - **Infinite** open-loop gain, $A \simeq \infty$
 - **Infinite** input resistance, $R_i \simeq \infty$
 - **Zero** output resistance, $R_o \simeq 0$

Ideal Op Amp in circuit analysis

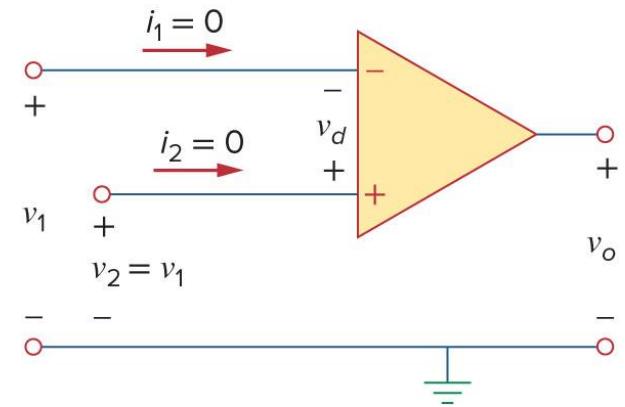
There are **two important properties** of Ideal Op Amp that are needed for analysing them in electric circuits.

1. The **currents** into both terminals are **zero**

$$i_1 = 0 \quad \& \quad i_2 = 0$$

2. The **voltage** across the input terminals is **zero**

$$v_d = v_2 - v_1 = 0 \quad \text{or} \quad v_1 = v_2$$



- Having $R_i \simeq \infty$ implies **open circuit** at the **input** terminals. Thus, **input currents** i_1 and i_2 should be **zero**.
- If $v_d \neq 0$, the output voltage $v_o = Av_d$ may become so high that it would exceed V_{cc} due to the large value of the open-loop gain A .

Note: $v_2 = v_1$ only happens when connecting the **output** to the **inverting** terminal (**negative feedback**).

Problem Solving in Op Amp Circuits

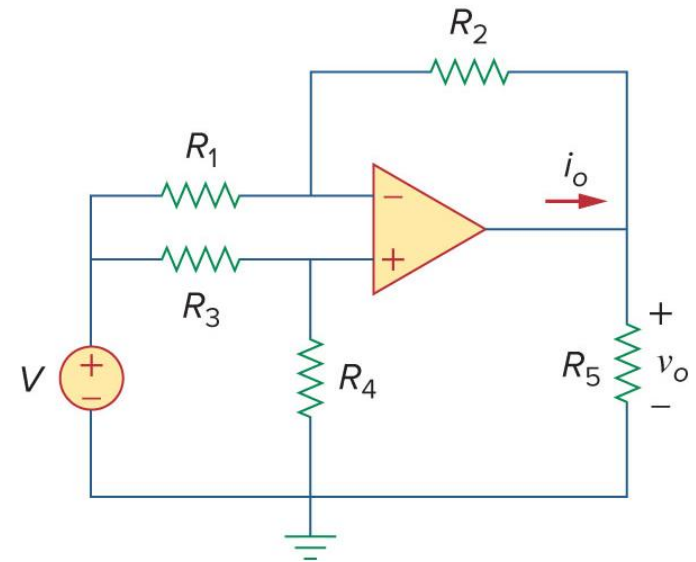
- Follow these three steps when solving Op Amp circuits:

1. Use **Ideal** Op Amp model:

- $i_1 = i_2 = 0$
- $v_1 = v_2$
- $R_i \simeq \infty$
- $R_o \simeq 0$

2. Apply **nodal** analysis.

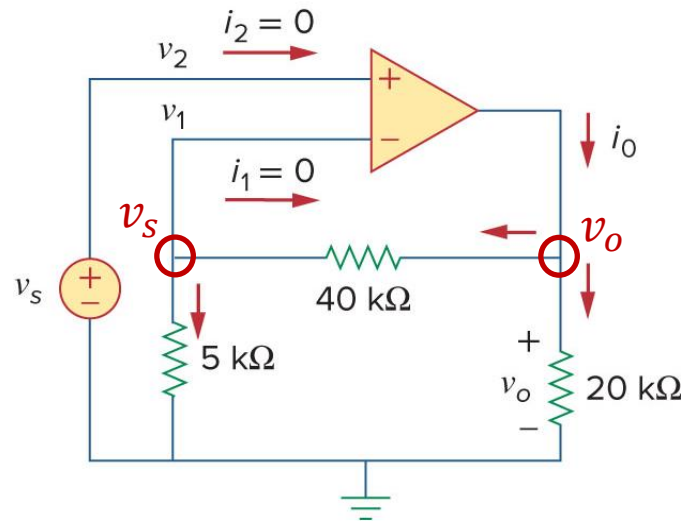
3. Solve nodal equations to express the **output voltage** in terms of **input signals** (voltage or current) or just solve for the **unknown** voltage or current in the circuit.



- Having $R_o \simeq 0$ implies **short circuit** at the **output** terminals, thus **output current** i_o can attain **any value** and **direction** depending on the circuit

Exercise 1

- a) Find the ratio of output voltage v_o to input voltage v_s , known as *closed-loop gain* $\frac{v_o}{v_s}$.
- b) Calculate i_o if $v_s = 1$ V.



Exercise 1

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- b) Calculate i_o if $v_s = 1$ V.

Ideal Op Amp: $v_1 = v_2 = v_s$

KCL at node v_s :
$$\frac{v_o - v_s}{40 \text{ k}} = \frac{v_s}{5 \text{ k}} + 0$$

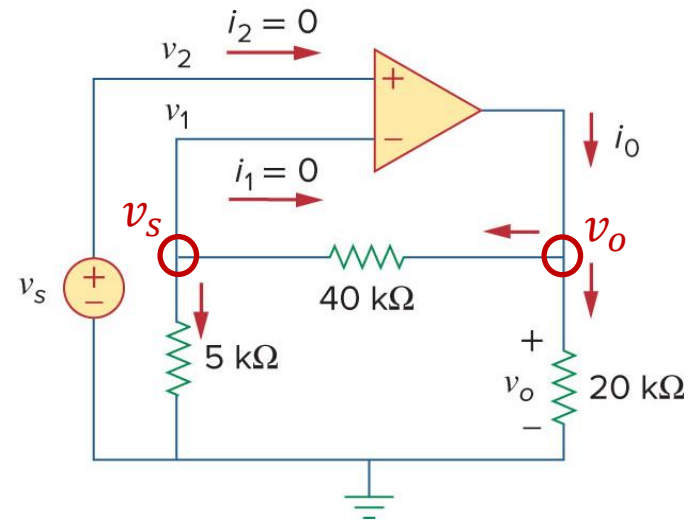
$$v_o - v_s = 8v_s \rightarrow v_o = 9v_s \Rightarrow \boxed{\frac{v_o}{v_s} = 9}$$

To find i_o , apply
KCL at node v_o :

$$i_o = \frac{v_o - v_s}{40 \text{ k}} + \frac{v_o}{20 \text{ k}}$$

If $v_s = 1$ V, $v_o = 9v_s = 9$ V, substitute these in node equation:

$$i_o = \frac{8}{40 \text{ k}} + \frac{9}{20 \text{ k}} = 0.2 + 0.45 = \boxed{0.65 \text{ mA}}$$



Inverting amplifier

- An **inverting amplifier reverses** the polarity of the input signal while **amplifying** it.
 - The **noninverting** input is **grounded**.
 - The **inverting** input is connected to the output via a **feedback resistor**, R_f .
 - The input voltage v_i is also connected to the inverting input via another resistor, R_1 .

Apply KCL at node 1: $i_1 = i_2 \Rightarrow \frac{v_i - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$

Ideal Op Amp with
negative feedback:

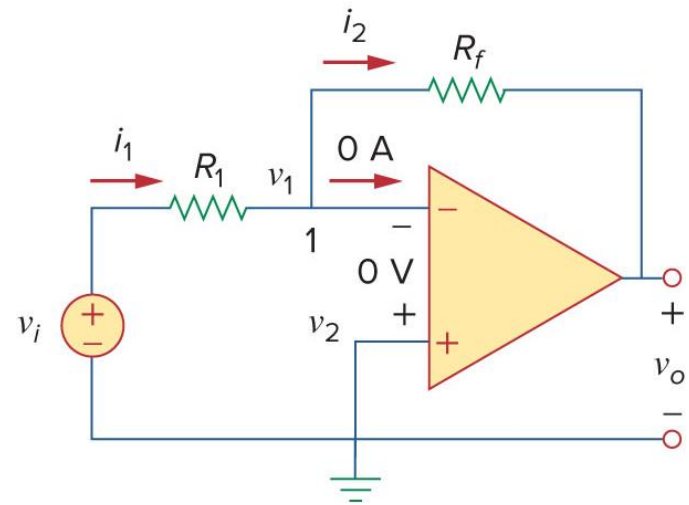
$$v_1 = v_2 = 0 \text{ V}$$

Thus:

$$\frac{v_i}{R_1} = -\frac{v_o}{R_f}$$

$$A_v = \frac{v_o}{v_i} = -\frac{R_f}{R_1} \quad (\text{voltage gain})$$

$$v_o = -\frac{R_f}{R_1} v_i$$



Non-inverting amplifier

- A **non-inverting amplifier** is designed to provide **positive** voltage gain amplification
 - The voltage input v_i is directly connected to **noninverting** input
 - The **inverting** input is **grounded** via R_1 and also connected to the output via the **feedback resistor**, R_f

Apply KCL at node 1: $i_1 = i_2 \Rightarrow \frac{0 - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$

Ideal Op Amp with negative feedback:

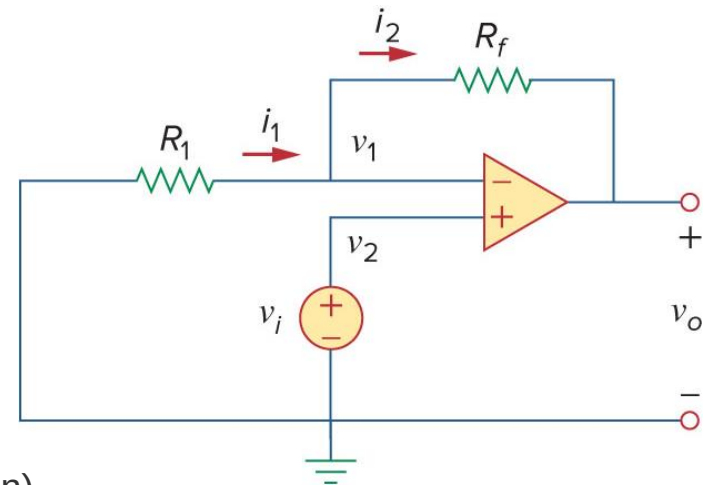
$$v_1 = v_2 = v_i$$

Thus:

$$\frac{-v_i}{R_1} = \frac{v_i - v_o}{R_f}$$

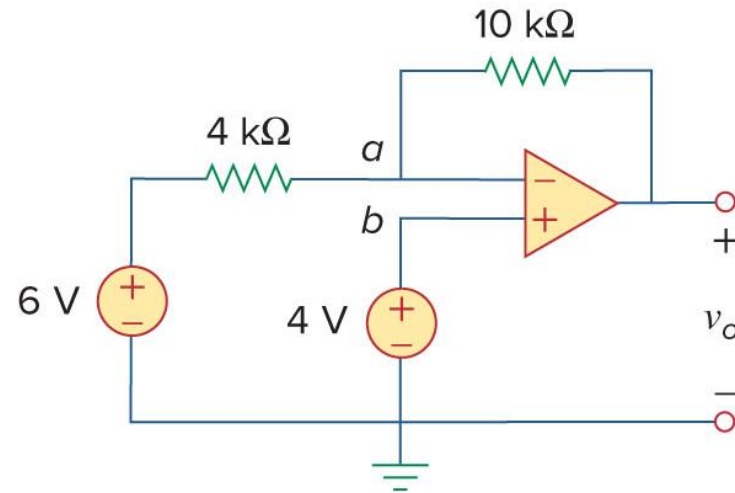
$$A_v = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_1} \quad (\text{voltage gain})$$

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$



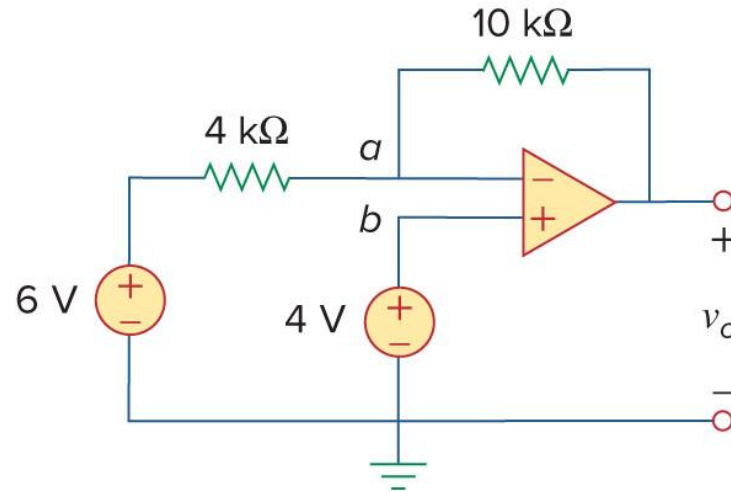
Exercise 2

Find the output voltage v_o in the Op Amp circuit shown below.



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Find the output voltage v_o in the Op Amp circuit shown below.



Using nodal analysis:

KCL at node a :

$$\frac{6 - v_a}{4} = \frac{v_a - v_o}{10}$$

Also, $v_a = v_b = 4$, and so:

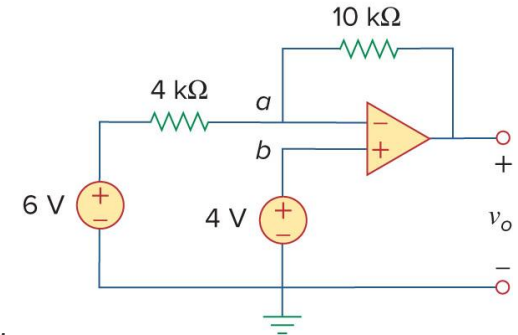
$$\frac{6 - 4}{4} = \frac{4 - v_o}{10} \Rightarrow 5 = 4 - v_o$$
$$v_o = -1 \text{ V}$$

Exercise 2

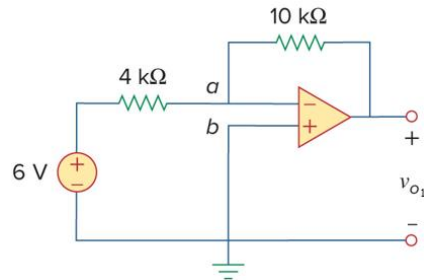
Find the output voltage v_o in the Op Amp circuit shown.

Using superposition: $v_o = v_{o1} + v_{o2}$

where v_{o1} is due to 6-V voltage source and v_{o2} is due to 4-V voltage source.

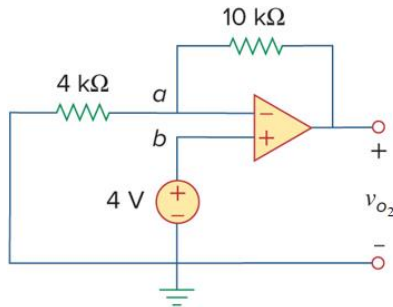


- For v_{o1} , set 4-V voltage source to zero and you will have an inverting Op Amp:



$$v_{o1} = -\frac{10}{4} \times 6 = -15 \text{ V}$$

- For v_{o2} , set 6-V voltage source to zero and you will have a non-inverting Op Amp:

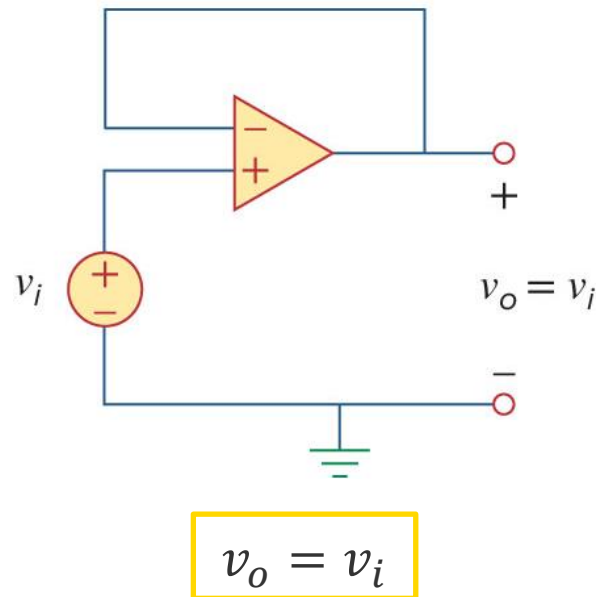


$$v_{o2} = \left(1 + \frac{10}{4}\right) \times 4 = 14 \text{ V}$$

Thus: $v_o = v_{o1} + v_{o2} = -15 + 14 = -1 \text{ V}$

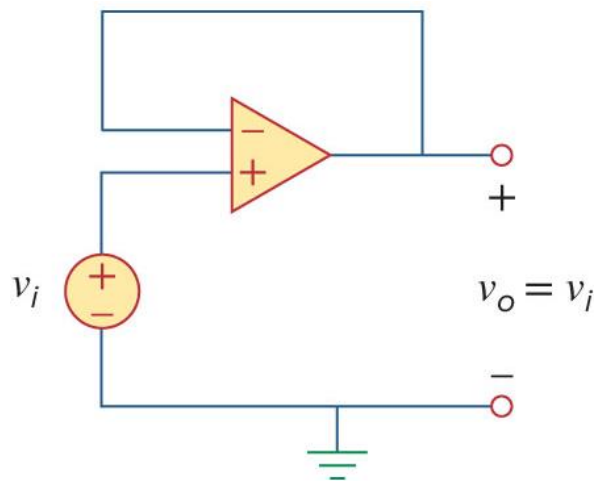
Voltage follower

- A **voltage follower** or *unity gain amplifier* is an Op Amp circuit whose **output** is the **same as the input** (i.e. gain equals 1).
 - Using Op Amp with negative feedback makes the **closed-loop gain** depend only on the **external resistors** rather than open-loop gain A .
 - A special case of non-inverting Op Amp is obtained when $R_f = 0$ or $R_1 = \infty$ or both. This makes the gain to become 1, i.e., $v_o = \left(1 + \frac{R_f}{R_1}\right) v_1 = v_1$.

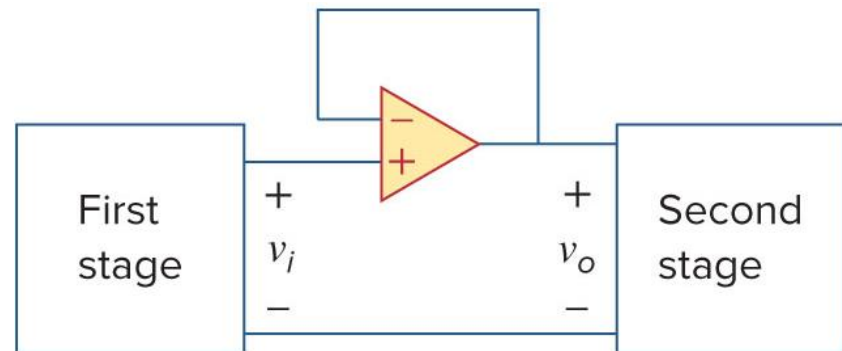


Voltage follower

- Voltage follower has a very **high input resistance** (impedance) and it is useful to **isolate** one circuit (first stage) from another (second stage).
- The second stage circuit **would not draw current** from the first stage, instead **Op Amp provides current** and power while maintaining the same voltage as v_i , **removing the loading effect** from the first stage circuit.



$$v_o = v_i$$



Summing amplifier

- A **summing amplifier** is an Op Amp circuit that **combines several inputs** and produces a **weighted sum** of the inputs in its output.
 - Aside from amplification, the Op Amp can be made to **sum** the values of input signals.
 - Using the **inverting Op Amp**, we can connect many inputs at the same time.

KCL at node a : $i = i_1 + i_2 + i_3$

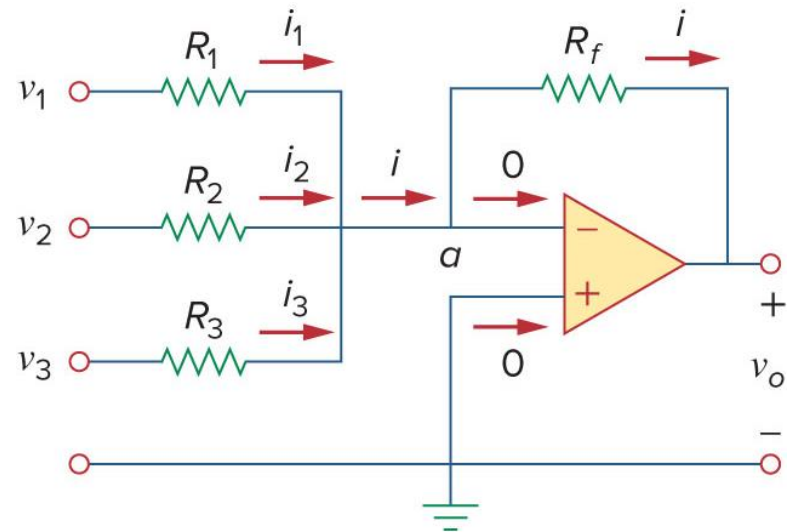
Ohm's Law:

$$\frac{v_a - v_o}{R_f} = \frac{v_1 - v_a}{R_1} + \frac{v_2 - v_a}{R_2} + \frac{v_3 - v_a}{R_3}$$

Ideal Op Amp with
negative feedback: $v_a = 0 \text{ V}$

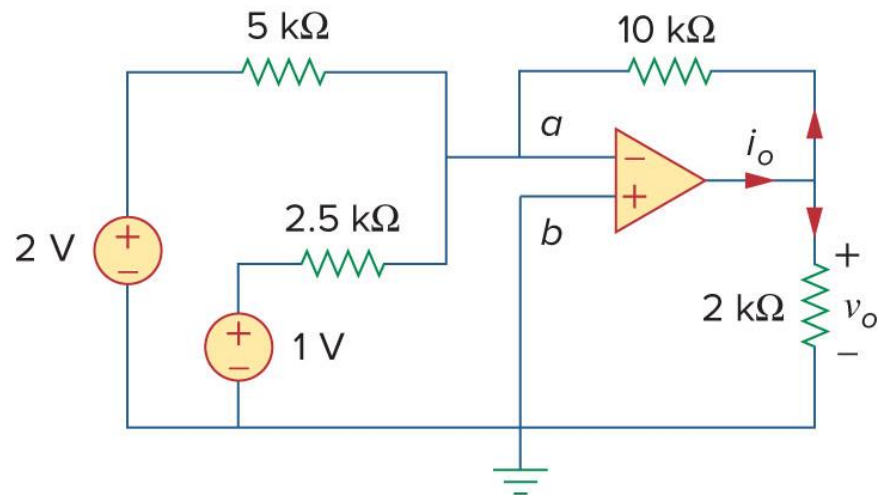
Thus,

$$v_o = - \left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right)$$



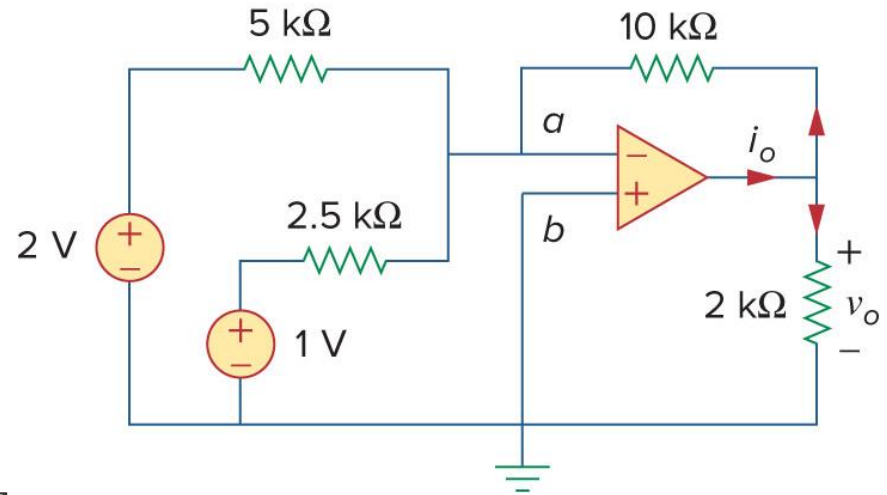
Exercise 3

Find the output voltage v_o and output current i_o in the Op Amp circuit shown.



Exercise 3

Find the output voltage v_o and output current i_o in the Op Amp circuit shown.



This is a summing amplifier with two inputs:

$$v_o = - \left[\frac{10}{5} \times 2 + \frac{10}{2.5} \times 1 \right] = -(4 + 4) = -8 \text{ V}$$

Applying KCL at the output:

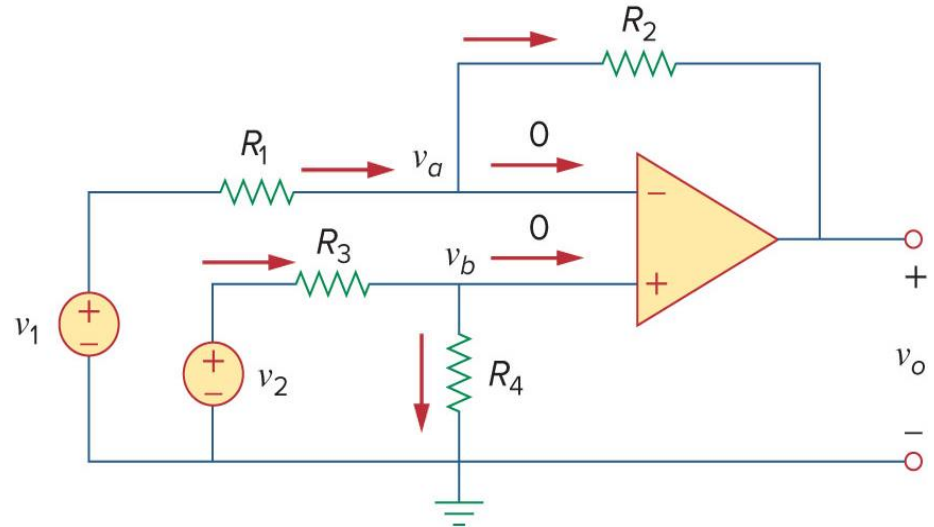
$$i_o = \frac{v_o - v_a}{10} + \frac{v_o - 0}{2} \text{ mA}$$

Ideal Op Amp with negative feedback makes $v_a = v_b = 0 \text{ V}$, thus:

$$i_o = \frac{v_o - 0}{10} + \frac{v_o}{2} = -0.8 - 4 = -4.8 \text{ mA}$$

Difference amplifier

- **Difference amplifiers** are used in various applications where there is a need to **amplify the difference** between two input signals.



KCL at node a : $\frac{v_1 - v_a}{R_1} = \frac{v_a - v_o}{R_2}$ or $v_o = \left(\frac{R_2}{R_1} + 1 \right) v_a - \frac{R_2}{R_1} v_1$ **I**

KCL at node b : $\frac{v_2 - v_b}{R_3} = \frac{v_b - 0}{R_4}$ or $v_b = \frac{R_4}{R_3 + R_4} v_2$ **II**

Ideal Op Amp with negative feedback makes $v_a = v_b$.
Substitute Eq. **II** into Eq **I**:

$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

A difference amplifier **must reject** a signal **common** to the two inputs. This condition exists if:

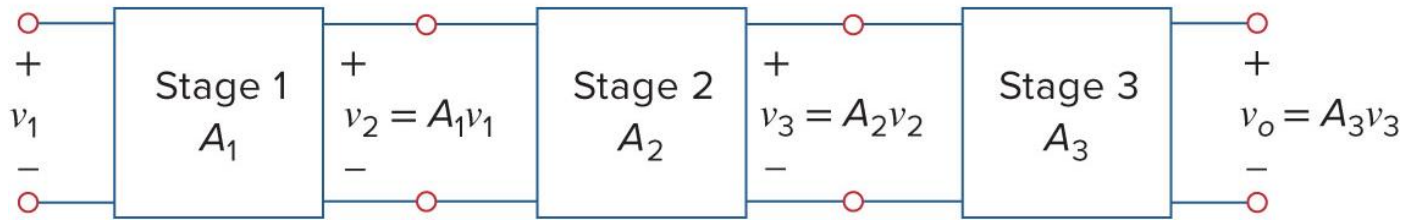
$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$



$$v_o = \frac{R_2}{R_1} (v_2 - v_1)$$

Cascade of Op Amp stages

- It is common to use multiple Op Amp circuits chained together to **increase** the **overall gain** of an amplifier.
 - Due to input and output impedances of the ideal Op Amps, stages can be connected together without affecting the performance of each other (**no loading effect**).
- This head to tail configuration is called “**cascading**”.
- Each amplifier is then called a “**stage**”.



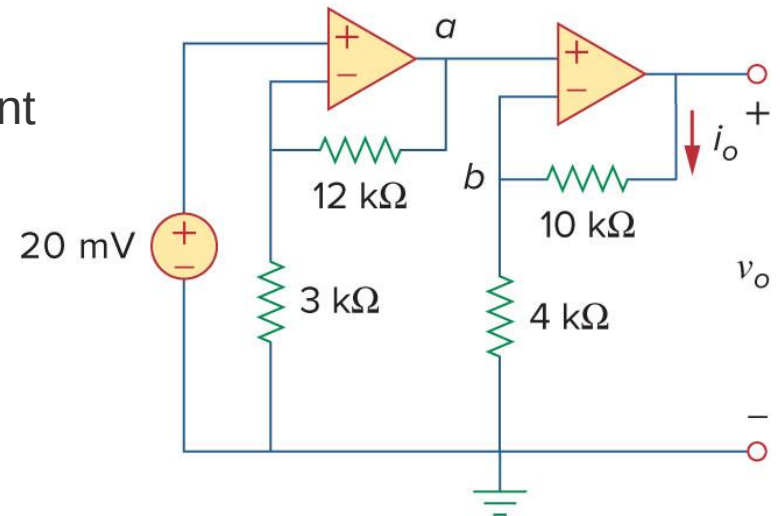
- The gain of a **series of amplifiers connected in cascade** is the **product** of the **individual gains**:

$$A = A_1 \times A_2 \times A_3$$

Note: The range of input voltage must be **small enough** to avoid **saturation** when amplified.

Exercise 4

Find the output voltage v_o and output current i_o in the Op Amp circuit shown.



Exercise 4

Find the output voltage v_o and output current i_o in the Op Amp circuit shown.

This circuit consists of two non-inverting amplifiers.

At the output of the 1st Op Amp:

$$v_a = \left(1 + \frac{12}{3}\right) \times 20 = 100 \text{ mV}$$

At the output of the 2nd Op Amp:

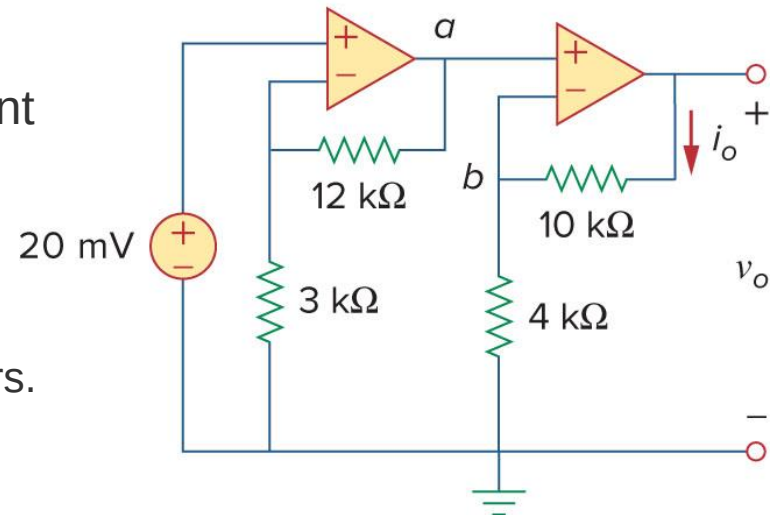
$$v_o = \left(1 + \frac{10}{4}\right) \times v_a = (1 + 2.5) \times 100 = 350 \text{ mV}$$

Current i_o through 10- Ω resistor:

$$i_o = \frac{v_o - v_b}{10} \text{ mA}$$

Ideal Op Amp with negative feedback makes $v_a = v_b = 100 \text{ mV}$:

$$i_o = \frac{(350 - 100) \times 10^{-3}}{10 \times 10^3} = 25 \mu\text{A}$$



Integrator

- **Capacitors** in combination with op-amps can be used to perform advanced mathematical functions.
- An **integrator** is an Op Amp circuit used to **integrate** signals.
 - By replacing the **feedback resistor** with a **capacitor**, the output voltage is obtained as follows.

KCL at node a : $i_R = i_C$

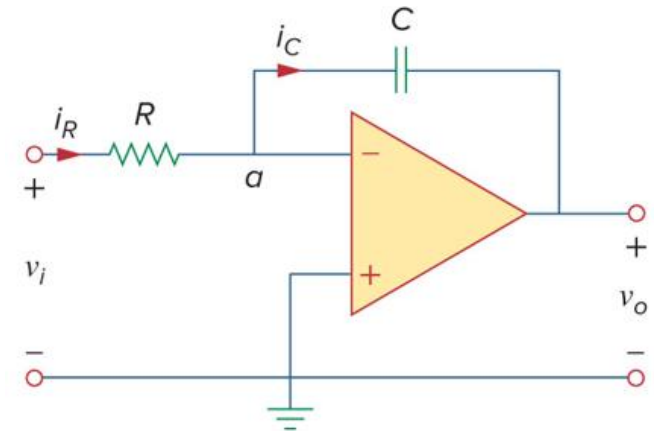
Current-voltage relations: $i_R = \frac{v_i}{R}$, $i_C = -C \frac{dv_o}{dt}$

Substitute in KCL: $dv_o = -\frac{1}{RC} v_i dt$

Integrate both sides: $v_o(t) - v_o(0) = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau$

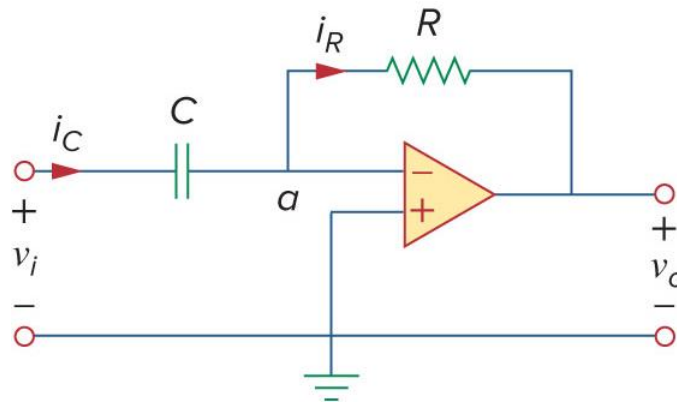
Ensuring zero initial conditions $v_o(0) = 0$ V:

$$v_o(t) = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau$$



Differentiator

- A **differentiator** is an Op Amp circuit used to **differentiate** signals.
 - If the **capacitor** is used in place of the **input resistor** instead of the feedback resistor, there will only be current flowing if the capacitor is changing.

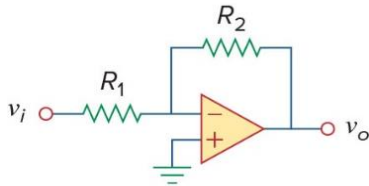


KCL at node a : $i_R = i_C$

Current-voltage relations: $i_R = -\frac{v_o}{R}$, $i_C = C \frac{dv_i}{dt}$

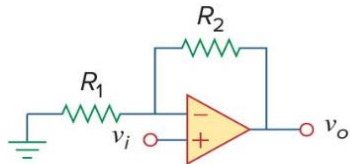
Substitute in KCL: $v_o(t) = -RC \frac{dv_i}{dt}$

Summary



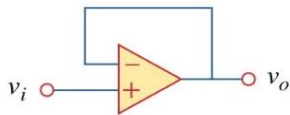
Inverting amplifier

$$v_o = -\frac{R_2}{R_1}v_i$$



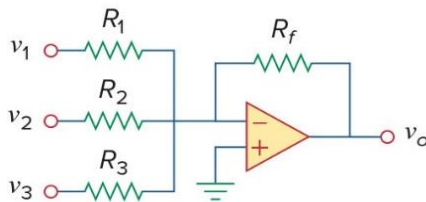
Noninverting amplifier

$$v_o = \left(1 + \frac{R_2}{R_1}\right)v_i$$



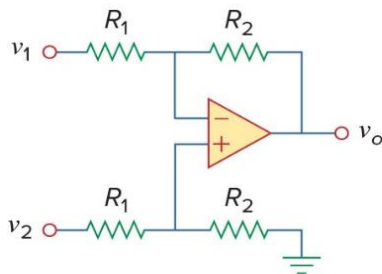
Voltage follower

$$v_o = v_i$$



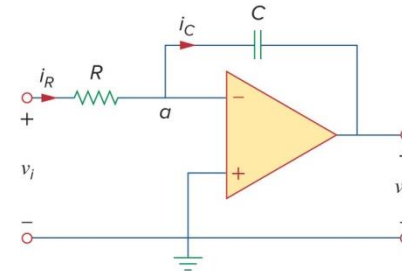
Summer

$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$



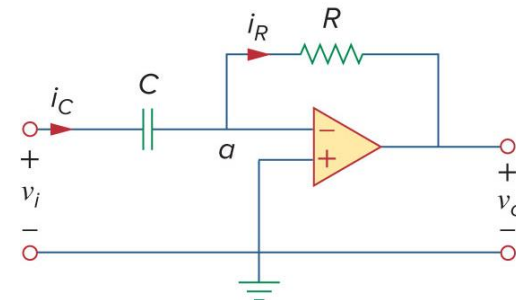
Difference amplifier

$$v_o = \frac{R_2}{R_1}(v_2 - v_1)$$



Integrator

$$v_o(t) = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau$$



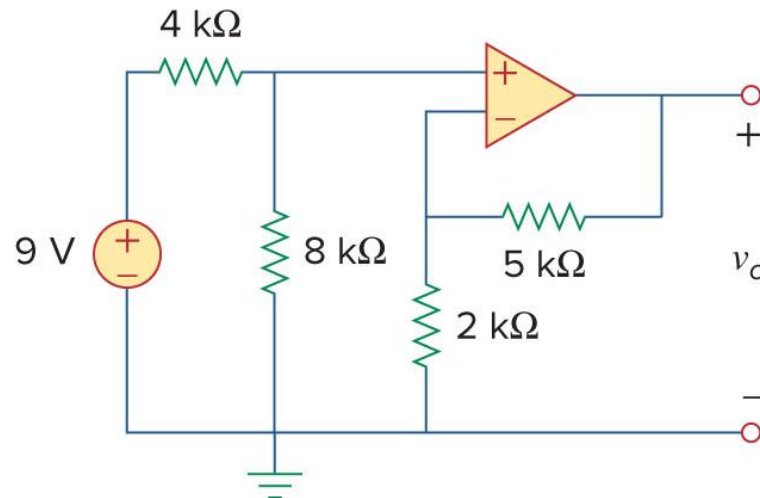
Differentiator

$$v_o(t) = -RC \frac{dv_i}{dt}$$

Exercise 5

Calculate the output voltage v_o in the Op Amp circuit below.

Solution in Exercises_solution_Topic_5.pdf



Questions?

